IIT Madras

ONLINE DEGREE

Machine Learning Techniques
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Gaussian Mixture Models

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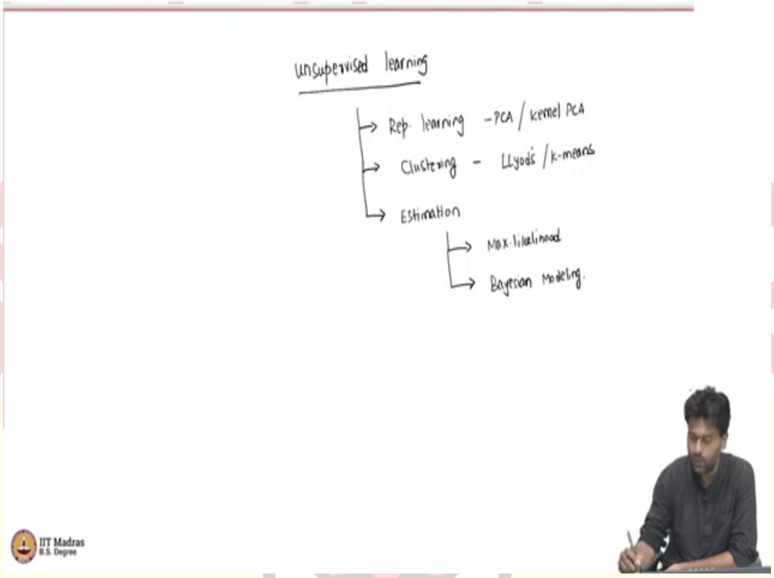

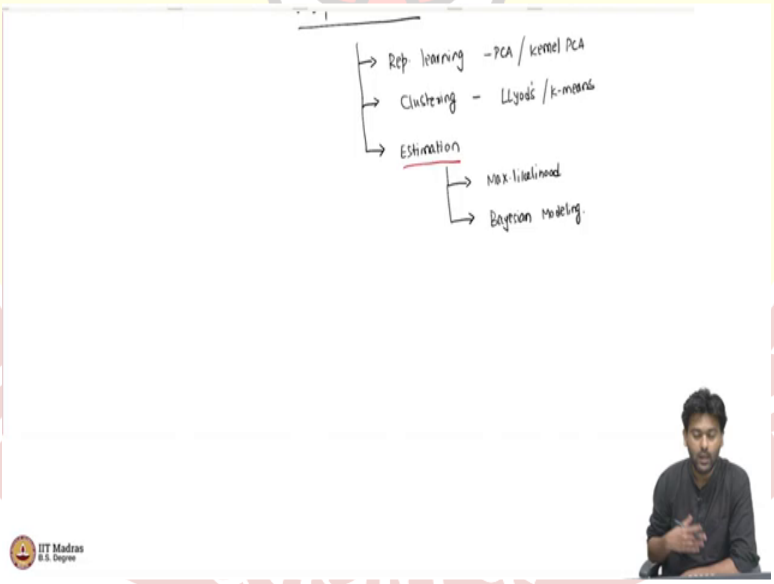

Welcome back, everybody. So, we have been looking at unsupervised learning so far in this course. One we will continue to look at unsupervised learning a little bit more. And just to recap, where we are in unsupervised learning, we looked at representation learning via the means of the PCA algorithm and also the kernel version of it, which we called as the kernel PCA. And then we also looked at clustering methods after that, which gave us the Lloyd's algorithm or the K-means algorithm.

And then, last time, we looked at estimation, as a probabilistic way of doing unsupervised learning. And in estimation, we looked at the maximum likelihood based ideas. And we also looked at a way to incorporate prior beliefs into our estimator, which is using the method of Bayesian modelling where you start with a prior and then you convert it into a posterior.

So, what we want to do today is continue our discussion about estimation in general. And in estimation, we want to look at a slightly more realistic type of data, which we would like to model in an unsupervised way.

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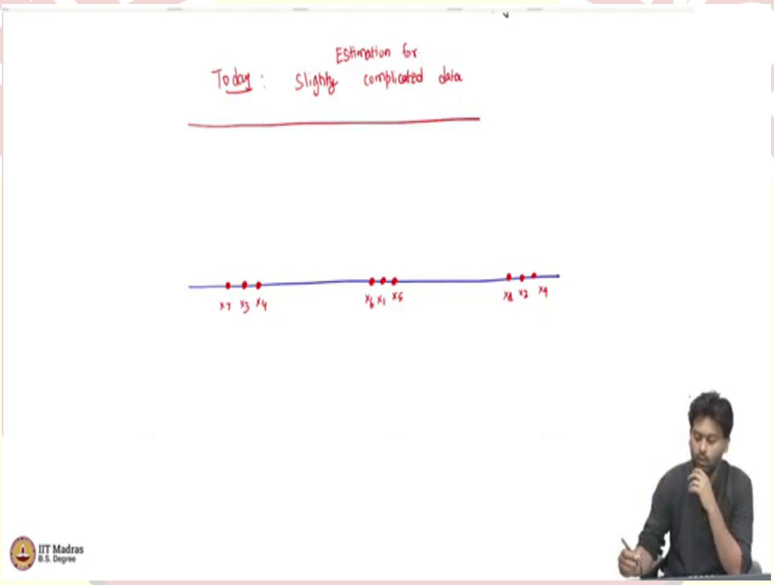

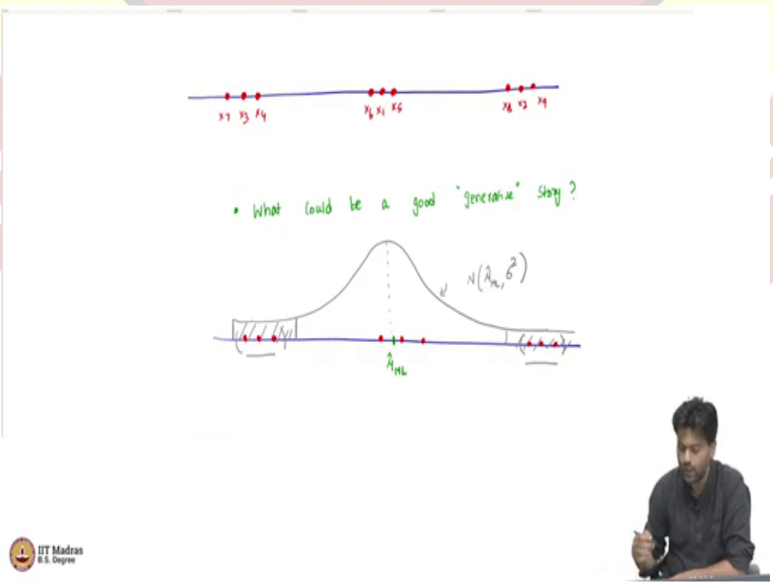

So, today, the goal is to look at slightly complicated data. And we look at estimation for this. So, what is this slightly complicated data that I am talking about? Again, what we will do is for illustrative purposes, we look at one dimensional data and try to understand the ideas but whatever I am going to talk today will carry forward for higher dimensions. In fact, whatever we discussed for estimation and maximum likelihood, Bayesian, everything works for high dimensional data also, but it is easiest to explain using one dimensional data.

So, what is the data that we are talking about here? So, let us say we are on the real line, simple one dimensional data and we have a bunch of data points, let us say like this. And the way the numbers are, the way these data points are ordered, let us say this is x1, this is x2, maybe x3 here, x4, I am just arbitrarily labeling these x8, x9. Let us say this is the data that we have. Now, in the world of estimation, the goal is you have to come up with a model that explains the data. So, the way we are thinking about this is, we will come up with a generative story that explains our data.

So, the question that will first ask is what could be a good generative story for this data? So, for this data, you see this data and then you ask the question, what could be a good generative data, a good generative story? First thing is this data is not just zeros and ones. So, we cannot use like a Bernoulli type of a modelling for here for this. This data has a bunch of real numbers.

So, one immediate thing that you could do is you can try measuring or modelling this using a Gaussian distribution, like how we did last time. So, because we have a bunch of real numbers, we can always use a Gaussian distribution to model it. So, I mean, we could model it. So, it is good or bad is we will talk about it, but then there is nothing stopping us from modelling it. And then you can apply your estimation methods and so on.

So, let us say if you did that. So, and then let us say we did a principle of maximum likelihood to get the best Gaussian, which has the mean, which is best in the sense that it explains the data in the best possible way in terms of maximizing the likelihood that we know that the mean of the maximum likelihood for Gaussian estimation is just the sample mean, which means in this case, the sample mean may be somewhere here.

So, this may be our $\hat{\mu}_{ML}$. So, the sample mean is the, this the maximum likelihood estimator, if you assume a Gaussian model. So, now if I want to generate new data from this model that I have learned, which has sample mean, $\hat{\mu}_{ML}$, let us say the variance is known in this case and the variance is some 1. Now, how would the density of the distribution look like with this particular sample mean that we have learned? Well, that is going to look something like this.

This is the Gaussian. Of course it the mean and the mode of the Gaussian match, it is going to be at the sample mean, because we are positing that the actual Gaussian that generated this data has to have a mean $\hat{\mu}_{ML}$. So, that is what we are estimating. And so if you look at the density, it is going to look like this.

Now the question is, well, what is this? This is the PDF of Gaussian with mean $\hat{\mu}_{ML}$ and variance some σ^2. Now, the question is, the more important question is, is this a good model for this data? Now, the problem with this model is, yes. So, the mean is the sample mean and which is somewhere in the middle of the data points, all that is fine. But then if you stare at this model for a while, you understand that there are these data points which have occurred in our data, but which has very less, which comes from very less dense regions.

So, if I try to find the probability of data coming in this region on the real line and in this region on the real line. Now, according to our hypothesis, the best Gaussian also has very less probability in these regions. However, we have seen data points in this region. So, now what is the problem? The problem is we have made an assumption that the model is Gaussian. And then maximum likelihood assumes that that is the gospel truth and then it is going to try to find the best mean, which explains the data under the model that we have assumed.

So, but if the model that we have assumed is not powerful enough to generate data of the form that we actually see, then maximum likelihood cannot do anything better. So, because it is searching in a space of Gaussians and then finding the best Gaussian. So, clearly we have seen a lot of data in these two regions, but then our explanation by via Gaussian is not very satisfactory, because these are low density regions for the Gaussian.

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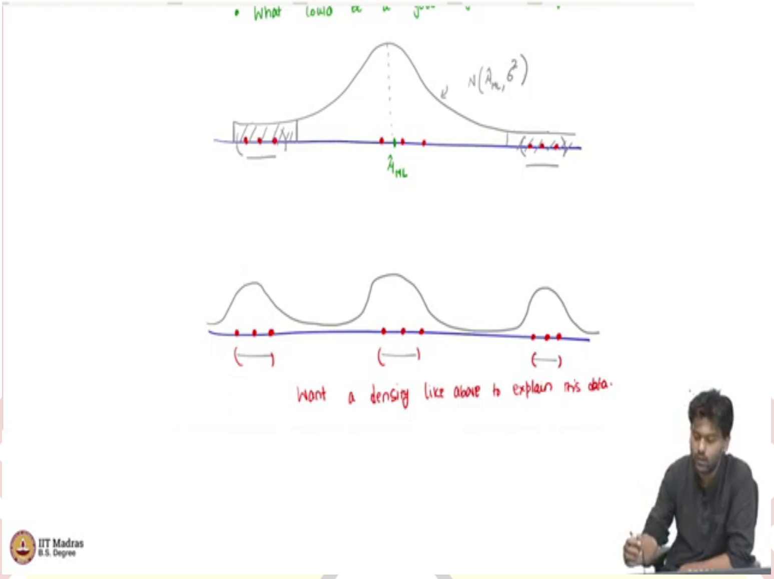

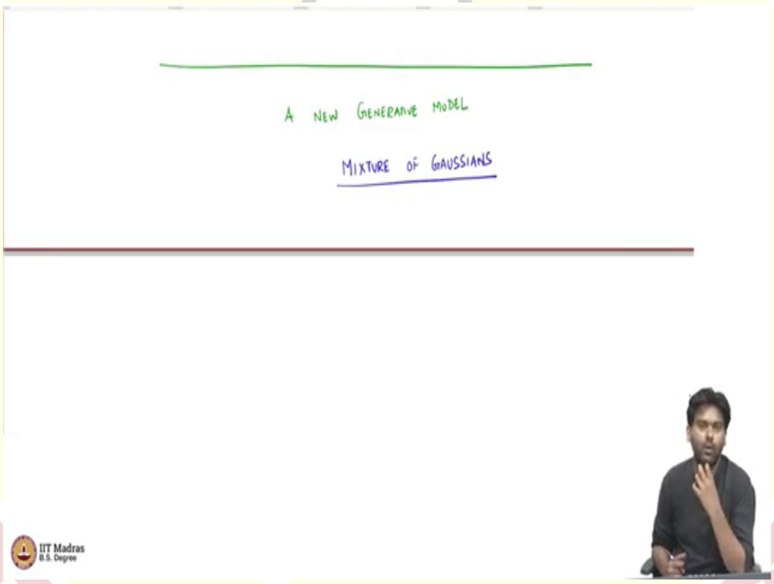

So, then how can we what do we want? Well, we want something like this. So, we do not want a Gaussian project to explain this data. But we want what kind of PDF do we want? So, if you think about this, I mean, this is a good place to pause and think, how do you think the shape of the PDF should be that explains this kind of data? What do you want? We want three different modes or three places where the density should peak.

So, there are some data points here, there are some data points here and there are some here. So, your PDF itself should have three modes. So, which means you need something like this perhaps. You need a density like this. So, let me write that down.

So, we want a density like above, to explain this data, why then a density like this has the property that it is it your data that you are actually seeing are coming from high density regions. So, perhaps this is a better model for this data. But then this is not a Gaussian. Clearly, this is not a Gaussian, we know Gaussians are unique model that is only one peak for a Gaussian, but then this needs three peaks. So, which means we need to come up with a different type of model to explain this type of data.

So, as you can see, there is some kind of clustering behavior in the dataset. So, that is what we will eventually get to. So, and then we want to somehow develop a probabilistic model, which can do this, which can model this kind of cluster data points. So, how can we do this? Well, each of these high dense region looks like a Gaussian in itself. But then overall, it is not a single Gaussian.

So, basically, what we then want is a new model. And what we want is a new generative model to explain this data, or the generative story and the name of this new model that we are going to come up is called as a mixture of Gaussians. It is not a single Gaussian. It is a mixture of Gaussians. In this case, it is a mixture of three Gaussians in the example that we just saw. So, what is the story? So, now what is the underlying story that generates this data?

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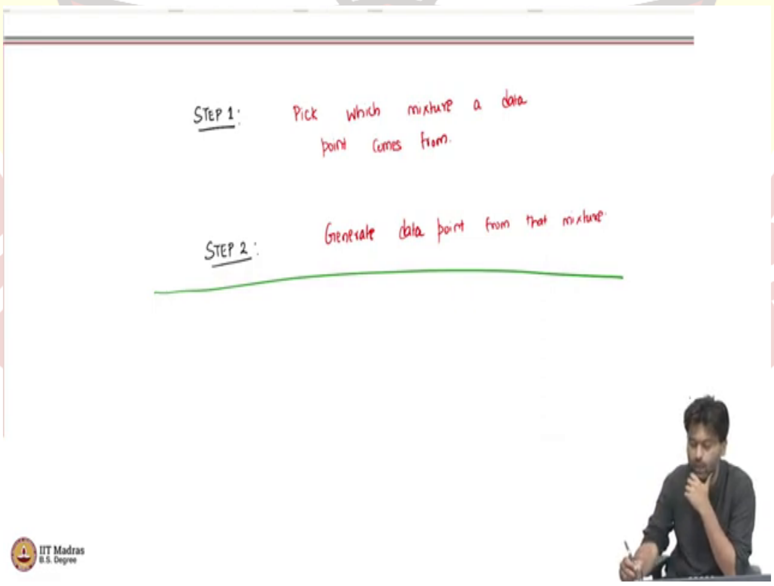

Whenever I say story, it means that what is the probabilistic mechanism that generates this data? So, I see the data. I need to understand. I need to put down a mechanism that generates the data. So, far in the estimation things that we have seen. The story is simple, you either sample from a Gaussian or you toss a coin into super simple. Now, such simple stories are not enough to explain this slightly complicated data. So, we need to come up with a new story. And the story that we will develop now has two steps.

I supposed to just tossing a coin, which is a single step, which will generate our data. And now we have a two step story. So, what is step 1 of the story? Well, we will look at both steps carefully, step 1 is the following.

So, pick which mixture, a data point comes from. So, the way we are going to model is. So, I have seen the data, I want to understand how each point is generated. I look at the first point. And now I want to explain how it was this point generated, x1, how was this generated? Well, before generating x1, something has happened in a probabilistic scenario. So, and that is what we are trying to explain now.

And that something has two steps to it. And the first step is when the mixture was first picked. So, somehow, we first decided, well, which of these mixtures there are three mixtures in the example that we showed. And let us for now assume that we know the number of mixtures. So, I give you a number of mixtures. And then you decide first somehow which mixture the data comes from, which you can think of it as mixture, or if you want to think of it as cluster, that is also fine. But which means the common parlance is mixture. So, I am going to stick to that.

So, which mixture the data comes from? So, once you have decided the mixture, then what do you do? Then it is pretty straightforward. So, then the second point is, once you have which mixture it comes from, well, you generate data from that mixture, generate data point from that mixture. So, this is a two step process, I supposed to the single step that we have seen so far. First, decide the mixture and then generate the data point from that mixture, super simple.

Now, of course, we need to make this more probabilistic. So, the way that I have just put down step 1 and step 2, there is no probabilities involved. But because it is a probabilistic model whose parameters we are trying to estimate, we need to make it more precise. And let us just do that.

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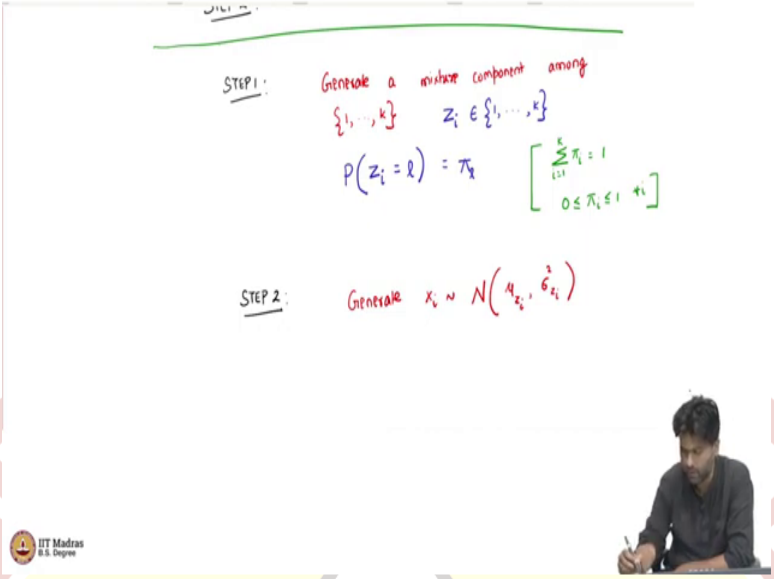

So, let us make this precise. So, step 1, in a more precise way is going to look like this. When I say we are figuring out which mixture this point comes from? Well, the way it is going to happen is imagine that, if you want to generate from a data point from three mixtures, then assume that the model has a dice, which has three faces. And now I am going to throw this dice. So, I am going to roll this dice and then the dice falls on one of the face. So, on that face, will have a number either 1, 2 or 3.

Now, we are going to think of this number as the mixture from which this data point is being generated. Let me formulate that. So, we are going to say we are going to generate a mixture component among, well, in the example it was 3 but general it can be k mixture. So, let us make it slightly more general, among 1 to k. There are k mixtures. And I am going to call this mixture component as zi. So, this is a number that we have to first decide between 1 to k, which indicates which mixture the ith data point comes from that values zi.

Now how am I deciding zi? I am deciding it in a probabilistic sense by rolling a dice. So, if the dice rolls and falls on the face 2, then it means that zi equals 2. The ith data point goes to the second mixture comes from the second mixture. So, what does that mean? That means that, there is a probability distribution, which is the dice that I am formulating it as probability distribution. The probability that zi equals let us use some other things. So, let us say l equals some πl. It just means that the probability that the ith data point comes from the lth mixture is given by some πl.

So, if there are only three mixtures, then let us say $\pi 1$ is 0.5, $\pi 2$ is 0.25, $\pi 3$ is 0.25, then it means that 50 percent of the time I am going to get a point from first mixture, 25 percent from second mixture and 25 percent from third mixture. So, we are assuming that this is step 1 of what is happening under the hoods. So, of course, π is a probability so which means that sum over i equals 1 to k πi will be 1. Because they are the sum of two probabilities, I need to choose one of the mixtures in a probabilistic way. And also, we know that 0 less than πi less than or equal to 1. Again, these are probabilities for all i.

All I am saying is that when you are selecting one mixture in a probabilistic fashion where the probabilities are given by $\pi 1$ to π capital K and which face the ith data points dice falls on, we are going to call that zi. So, if you remember from K means we also use zi as a cluster indicator for the data point here it is the mixture indicator, if you will. So, this is first step. So, now we have decided which cluster or which mixture the data has to come from.

Now the second point, second thing is second step is pretty simple. So, let us say zi was some 5, which means that I need to generate from data from the fifth mixture. Now for each mixture, we have a Gaussian with its own mean and variance. That is the assumption. So, every mixture has its own mean and variance. And now if I roll the dice, it falls on face 5, then it is as if I am going into the fifth door, which tells me there is a mean and variance sitting inside the fifth door and then I will pick a random data point according to that particular Gaussian.

To formalize, this, we will just say generate xi, the ith data point as a Gaussian or normal, which I am using N to denote with what is the mean? Well, the mean is the mean corresponding to μ_{zi}. So, zi is the cluster indicator, which door I am going in to pick a point. So, there is each mixture as a door and then let us say go into the fifth door, then means zi was 5. That is why I went to the fifth floor. And then I am sampling a point according to the mean and $\sigma^2 zi$ according to that particular mixture.

So, this is the generative model, it has two steps. And this is how one data point is generated. To generate a single data point, you go through these two steps. First roll a dice to cut which mixture and then go to that mixture sample Gaussian from that Gaussian data point according to the mean and variance of that mixture.

Now, to generate the second point, you again assume the same story. So, you again, roll the dice, it might fall on a different face, you go to that corresponding mixture and then sample and keep doing this and different times you get a dataset.

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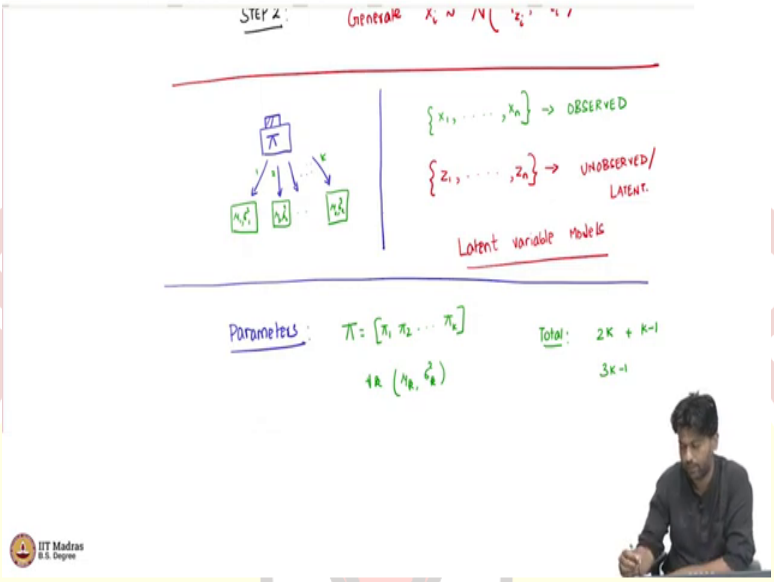

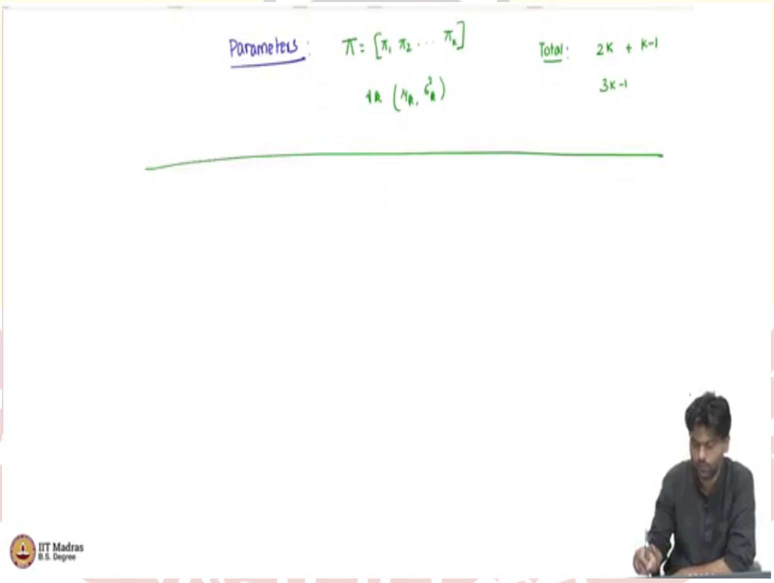

So, if you have to put this in the in pictorial form, like how we have been seen so far, it will look something like this. So, you have the first box, which has π sitting in it. You press this button, what you are going to get is well, one of the values which is 1 to k with different probabilities determined by π. So, once you decide which box, then you go to that particular box and then each of these 1 to k has a corresponding box, which has $\mu1$ σ^21, $\mu2$ σ^22,..., μk σ^2k.

Now and then, according to each of this, when depending on this, let us say zi was 2, I went to the second box, I get xi from that and so on. So, that is in general. So, this let it be general. So, now, this is a slightly different model from what we have seen so far, because it has two steps, which means of course, we have the data x1 to xn, which we are observing, which are real numbers and these are the observed quantities.

But then in the model that we are put down, there are some unobserved quantities also, that determines x1 to xn. And these in this particular case, what are the unobserved quantities? Well, the unobserved quantities are z1 to zn, the mixture indicator or the cluster indicator or unobserved. So, these are unobserved or latent. So, these are what are called as latent variable models.

So, our Gaussian mixture model is a latent variable model, because the final output that you are observing, you are assuming depends not only on some parameters that you want to estimate, but also on some unobserved latent variables. So, these are latent variable models. It is just an example. There are several latent variable models that people typically use and this is one of the most commonly used ones.

So, now as an estimation procedure, the question is, what are the parameters that we need to estimate in this model? This is a good time to pause and think how many parameters will determine this model completely? I will tell you that now. So, what are the parameters? You only see the data now, what are the parameters that you need to figure out? Well, in the Gaussian case, simple Gaussian case, it was just the mean or mean and variance.

Now we are seeing there are many more parameters. So, I mean, the first thing is there is a π, which is a vector of probabilities, π1, π2,..., πK. This is something that we do not know, but then it determines the output. So, this is a parameter of interest. And then for each K there is a μK and σ^2K. So, there are two parameters. In total, you have 2K mean and variance for each of the mixtures. That is that is 2K parameters, plus there is a common π sitting well, which gives probabilities for each data points.

You can either think of π as having K values, but then because π has this condition that the sum of πi's should be 1. If you know K - 1 of them, the last one comes for free. So, because they have to summed one, there is a there is a restriction there. So, you can think of it as K - 1 free parameters, if you will. So, overall, it is of order of 3K parameters. I mean, if you want to be pedantic, you can say it is 3K - 1 that is also fine. So, we need to estimate these many parameters from data.

So we are earlier, we were just estimating one parameter, the bias of the coin or the mean of the Gaussian distribution. Now, there are host of other parameters that we want to estimate. Now, it is not obvious what are these estimators? So, just by looking at the data, how can we decide what are πs, what are μs and σ^2? It is not at all clear. So, it is hard to make a educated guess as to what these things would be.

But our method of maximum likelihood is a very principled method that does not that will give you an answer no matter what the model is going to be. So, as long as we have a well specified probabilistic model, we know that you can run the principle of maximum likelihood on it. So, let us try that. So, that is what we are going to do next.